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DE Assignment - 5

Q1 The 8085 is a 40-pin microprocessor. Its pins are divided into different functional groups based on their operations.

1. Address Bus
   - Carries 16-bit address (A0-A15)
   - Unidirectional

2. Data Bus
   - Carries 8-bit data (D0-D7)
   - Bidirectional

3. Control & Status Signals
   - Controls internals and external operations
   - Example: RD, WR, ALE

4. Serial Input/Output Signal
   - Used for serial communication
   - SID - Serial Input Data
   - SOD - Serial Output Data

5. Externally Initiated Signal
   - Used for interrupts and DMA
   - Eg. INTR, RST 7.5, Hold

6. Power Supply & Clock Signal
   - Provide power and timings
   - Pins: Vcc, Vss, X1, X2

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Q2 Addressing Mode of 8085

1. Registering Addressing Mode
   - Operand is stored in a register
   - Example : MOV A, B

2. Immediate Addressing Mode
   - Data is part of the instruction itself
   - Example : MVI A, 47H

3. Direct Addressing Mode
   - A 16-bit memory address is given directly
   - Example : STA 3050 - H

4. Indirect Addressing Mode
   - Addressing is stored in a register pair
   - Example : MOV A, M

5. Implied Addressing Mode
   - Operand is implied in the instruction
   - Example : CMA, HLT

Q3 8085 Instruction Set Classification:

1. Data Transfer Instructions
   - Transfer data between registers and memory
   - Examples: MOV, MVI, LDA, STA

2. Arithmetic Instruction
   - Perform arithmetic operations
   Examples ADD, SUB, INR, DCR

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3. Logical Instructions
   - Perform logical operations
   - Examples: ANA, ORA, XRA, CMP

4. Branch Instructions
   - Change program sequence
   - Example: JMP, JZ, CALL, RET

5. Control Instructions
   - Control processor operations
   - Examples: HLT, NOP, EI, DI

Q4 Register Organization of 8085

1. General Purpose Register (8-bit)
   - Register: B, C, D, E, H, L
   - Can be combined as register pairs BC, DE, HL

2. Accumulator (A)
   - Main register for arithmetic and logical operations

3. Program Counter (PC)
   - 16-bit register
   - Stores address of the next instruction

4. Stack Pointer (SP)
   - 16-bit register
   - Point to the top of the stack

5. Flag Register (F)
   - Stores the status of operation

6. Internal Register
   - Temporary register
   - Instruction register

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Q: Flag Register & Significance of Each Flag

The flag register is an 8-bit register that contains 5 - status flag, which indicate the result of operation.

Flags

- Sign flag (S): set if result is negative
- Zero flag (Z): Set if result is 0 zero
- ~~Auxiliary~~ Auxiliary Carry (AC): Carry from bit 3 to bit 4.
- Parity flag : set if even number of 1s in result
- Carry flag : set if carry or borrow occurs.